Original research

Commensal *Bacteroides* T6SS alleviate GI-aGVHD via mediating gut microbiota composition and bile acids metabolism

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ABSTRACT

Background Gastrointestinal acute graft-versus-host disease (GI-aGVHD) is one of the main complications of patients undergoing allogenic haematopoietic stem cell transplantation (allo-HSCT). A deeper understanding of the mechanisms of sustaining intestinal homeostasis is essential.

Objective Here, we investigated micro-organisms and microbial metabolites that were crucial for intestinal homeostasis in the context of GI-aGVHD management.

Design We profiled the gut microbiota, immune indices and gut metabolism of 71 patients undergoing allo-HSCT. Initially, we set up a mouse aGVHD model to confirm the effect of *Bacteroides fragilis* type VI secretion system (T6SS) on aGVHD progression. Subsequently, we applied 16S amplicon sequencing and metabolic profiling to reveal the function of *B. fragilis* T6SS on microbial structure intestinal and metabolome. Finally, the mediation package was used to validate our findings in clinical samples.

Results A higher abundance of *Bacteroides* spp contributes to reducing the incidence of GI-aGVHD, and the T6SS is required for *Bacteroides* spp protection on aGVHD. T6SS-mediated antagonism regulates the structure and composition of gut microbiota, affecting the entire gut metabolome, particularly the bile acids metabolism, subsequently reducing inflammation response in the intestine and protecting intestinal barrier integrity. Notably, accumulating primary bile acids such as chenodeoxycholic acid exacerbated aGVHD by enhancing the activation of T cells. Mediation analysis further validated that T6SS affects the incidence of GI-aGVHD through its effect on primary bile acid metabolism.

Conclusions T6SS in commensal bacteria could modulate bile acid metabolism, potentially impacting aGVHD outcomes and offering a novel target for therapeutic interventions.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ The dysbiosis of the microbiota and the alterations of microbial metabolites are pronounced in acute graft-versus-host disease (aGVHD) patients and associated with the poor patient prognosis of patients. The microbiota interacts with the host by microbiota themselves or via microbial metabolites to regulate intestinal homeostasis and the immune response.
⇒ Faecal microbiota transplant, or targeted probiotic supplementation, has been proposed as a clinical strategy to restore intestinal homeostasis in aGVHD management.
⇒ Microbial intervention strategies require a deeper understanding of the interaction between gut micro-organisms in immunocompromised patients.

WHAT THIS STUDY ADDS

⇒ Type VI secretion system (T6SS) of *Bacteroides fragilis* is a key regulator in limiting T cell activation and promoting intestinal barrier via affecting gut microbiome and metabolome.
⇒ Accumulating primary bile acids (BAs) such as chenodeoxycholic acid exacerbated aGVHD by enhancing the activation of T cells.
⇒ Modulating primary BAs could enhance the role of T6SS gene families in reducing the risk of gastrointestinal (GI)-aGVHD.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Construct engineered T6SS with enhanced killing capacity to eliminate harmful bacteria.
⇒ Characterise bacteria-derived novel BAs for GI-aGVHD intervention.

INTRODUCTION

Allogenic haematopoietic stem cell transplantation (allo-HSCT) is currently the primary method in treating malignant haematological disorders such as leukaemia, lymphoma and aplastic anaemia.[1] Despite its curative potential, allo-HSCT is frequently complicated by the onset of acute graft-versus-host disease (aGVHD), a severe immune-mediated condition where donor immune cells recognise the recipient tissues as foreign and launch an attack. With an incidence rate ranging from 30%–70%, aGVHD remains one of the leading causes of non-relapse mortality following transplantation.[2] Among its various forms, gastrointestinal aGVHD (GI-aGVHD), which occurs in

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the lower GI tract, is strongly associated with poor outcomes, as the disruption of the intestinal mucosal barrier leads to increased interactions between bacterial products and immune cells, thereby exacerbating tissue damage and inflammation.[3 4] Emerging evidence highlights the critical role of the gut microbiota in the development and progression of aGVHD. Over the course of allo-HSCT, patients showed profound shifts in microbial communities marked by a significant reduction in overall diversity. This shift was particularly prominent in patients who subsequently developed aGVHD. Increases in *Lactobacillales* and decreases in *Clostridiales* also happened after allo-HSCT for aGVHD subjects.[5] Independent studies conducted by different centres have reported the efficacy of faecal microbiota transplant (FMT) in restoring intestinal bacterial diversity, suggesting that targeted supplementation, aimed at modulating the intestinal flora through probiotics or beneficial bacterial administration, could be a promising strategy for managing aGVHD.[6–9] Recent studies have made further efforts to distinguish beneficial bacteria and bacteria-derived metabolites in aGVHD management.[10] Bacterial metabolites like short chain fatty acids could have a protective effect by reducing inflammation responses and preserving barrier integrity. Nevertheless, significant gaps remain in our understanding of the complex interactions between the intestinal microbiota and host immune responses.

In the competitive environment of the gut, where nutrients and space are limited, the intestinal microbiota emerges as an antagonistic microbial community. Opposing competitors can be isolated from each other by buffer zones, thereby reducing the extent of interspecies interactions,[11] which, in turn, maintains a long-term stable community. Multiple protein secretion systems in the intestinal flora mediate competition between bacteria to maintain composition and abundance, including T3SS, T4SS, CDI, T6SS, LXG proteins and OME.[12] The involvement of these antagonistic systems in intestinal homeostasis, especially in relation to disease development, is gradually being revealed. Among them, T3SS genes were identified as biomarkers for diagnosing and monitoring the clinical status of Crohn's disease patients.[13] Notably, genomic studies reveal that approximately 50% of *Bacteroides* species possess type VI secretion system (T6SS), which deploys a syringe-like apparatus to inject toxic effectors into target cells.[14 15] In fact, the T6SS has garnered significant attention due to its dual role in interbacterial competition and its ability to modulate eukaryotic cell functions.[16–19] It is estimated that more than 10^9 T6SS secretory events occur per gram of colon per minute, demonstrating the critical role of T6SS in regulating and shaping intestinal homeostasis.[20]

This study systematically analysed the microbiota, immune indices, metagenomic profiles and metabolic alterations in patients with GI-aGVHD. Our findings demonstrated that *Bacteroides* T6SS significantly influenced the gut microbial composition and further impacted the intestinal metabolome, particularly the bile acid (BA) metabolism. In mouse models, the accumulation of primary BAs further aggravated aGVHD, suggesting that T6SS-BA metabolism is a novel clinical intervention target for aGVHD treatment.

MATERIALS AND METHODS

We profiled the gut microbiota, immune indices and gut metabolism of 71 patients undergoing allo-HSCT. Mouse aGVHD model was constructed to confirm the effect of *Bacteroides fragilis (B. fragilis)* T6SS on aGVHD progression. Recipient mice were euthanised 14 days after bone marrow transplantation (BMT). The spleens were collected for flow cytometry analysis. The liver and lung tissues were collected for histopathology analysis. The intestine tissues were collected for flow cytometry, histopathology, immunofluorescence, qPCR and cytokine analysis. Subsequently, we applied 16S amplicon sequencing and metabolic profiling to reveal the function of *B. fragilis* T6SS on microbial structure intestinal and metabolome. Finally, the mediation package was used to validate our findings in clinical samples. The detailed methods and materials (online supplemental tables S1 and S2) are provided in online supplemental files.

RESULTS

Bacteroides bacteria help to alleviate aGVHD

Faecal samples of 71 patients undergoing allo-HSCT were collected in this study, including 24 GI-aGVHD patients (GI-aGVHD group) and 47 non-aGVHD patients (non-aGVHD group) (figure 1A). The baseline clinical and transplantation characteristics of patients in the GI-aGVHD and non-aGVHD groups are shown in online supplemental table S3. No significant differences exist between the two groups in gender, age, donor type, donor gender, infused cells, conditioning regimen or aGVHD prevention regimen. At 4–6 weeks post-HSCT, coinciding with the onset of aGVHD (mean 31 days), a higher percentage of activated $CD4^+$ T cells in peripheral blood was observed in GI-aGVHD groups compared with the non-aGVHD group (figure 1B). However, the percentages of T, B and NK cells had no significant differences between GI-aGVHD and non-aGVHD groups (online supplemental figure S1A). Serum cytokines, including IL-6, IL-8, TNF-α, IFN-α2, CX3CL1, CXCL12, sST2, sTNF-RI, sTNF-RII and sTREM-1, were also elevated in the GI-aGVHD group compared with non-aGVHD group (figure 1C). 16S rDNA analysis revealed a dramatic reduction of overall microbiome richness (Chao1 diversity index) at 4–6 weeks post-allo-HSCT in the GI-aGVHD group compared with the non-aGVHD group (Wilcoxon rank sum test, figure 1D). In addition, our data showed a statistically significant association between lower baseline alpha diversity and increased aGVHD risk, as well as reduced survival (online supplemental figure S1B). Accordingly, PCoA plot analysis based on binary Jaccard distance and ternary plot for the top 10 taxa showed distinct microbiota structures between the two groups (figure 1E,F). Moreover, there was no impact of demographic/clinical variables on microbial diversity (online supplemental table S4). Spearman correlation analysis unravelled close interactions between altered microbiota taxa with inflammatory cytokines (figure 1G). The relative abundance of *Bacteroides* was negatively associated with the secretion of IL-6 and IL-8. These results indicated that gut microbiota might affect the progression of aGVHD by regulating the immune response.

For human faecal samples, significant differences in taxonomic compositions were determined via LEfSe analysis, with LDA score >3.5 and $p<0.05$ (figure 1H). At the genus level, the relative abundance of multiple bacteria varied significantly at 4–6 weeks post-HSCT (online supplemental figure S1C). Notably, the relative abundances of *Bacteroides*, *Roseburia* and *Faecalibacterium* were significantly lower in GI-aGVHD patients than in non-aGVHD controls (figure 1I and online supplemental figure S1C), although their associations with clinical outcomes differed. Higher abundance of *Bacteroides* was associated with reduced aGVHD incidence, improved overall survival (OS), and GVHD-free, relapse-free survival (GRFS) (figure 1J–L). Elevated *Roseburia* abundance correlated with lower aGVHD risk and improved

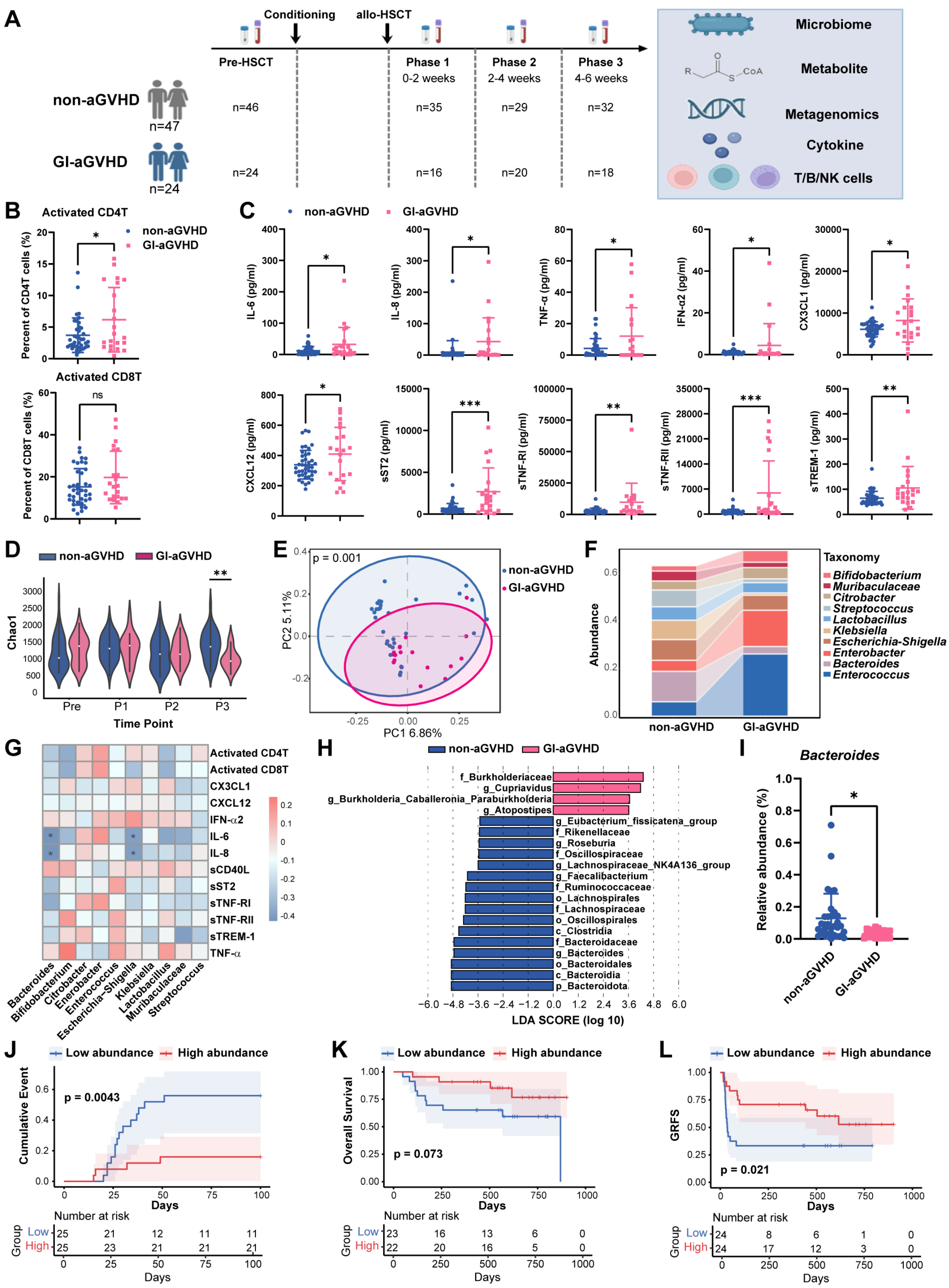

Figure 1 *Bacteroides* bacteria help to alleviate aGVHD. (A) Flow chart of patient enrolment and treatment. (B) Percentage of activated CD4^{+} and CD8 T^{+} cells in PBMCs of patients at 4–6 weeks post-allo-HSCT. (C) Level of inflammation cytokines in plasma of patients at 4–6 weeks post-allo-HSCT. (D) Alpha diversity of faecal samples from enrolled patients across treatment. (E, F) Principal coordinate analysis and Ternary plot for the top 10 taxa faecal samples from enrolled patients at 4–6 weeks post-allo-HSCT. (G) Spearman correlation analysis of the altered microbiota taxa with measured cytokines and T-cell populations. (H) LEfSe analysis on significantly different taxonomic compositions at 4–6 weeks post-allo-HSCT. (I) Relative abundance of *Bacteroides* genus at 4–6 weeks post-allo-HSCT. (J–L) KM curves of *Bacteroides* abundance on aGVHD risk, overall survival, and GRFS. Data are represented as mean±SD. Alpha diversity was analysed with the Wilcoxon rank sum test. PCoA plot based on binary Jaccard distance was generated using QIIME1, and the statistically significant separation was generated according to the PERMANOVA test. Each dot represents a single biological replicate (B, C, I). Survival curves were analysed with the log-rank test. *p<0.05, **p<0.01, ***p<0.001. aGVHD, acute graft-versus-host disease; GI, gastrointestinal; GRFS, GVHD-free, relapse-free survival; HSCT, haematopoietic stem cell transplantation; PBMCs, peripheral blood mononuclear cells; PCoA, principal coordinates analysis; PERMANOVA, permutational multivariate analysis of variance.

GRFS, though not with OS. In contrast, *Faecalibacterium* did not show protective associations with aGVHD incidence, OS or GRFS (online supplemental figure S1D, E). Additional murine aGVHD experiments confirmed that oral administration of *B. fragilis* and *Roseburia hominis* alleviated disease severity and improved survival, supporting their protective role in GVHD progression (online supplemental figure S1F, G).

T6SS is essential for *B. fragilis* protection in mouse aGVHD

T6SS is prevalent in *Bacteroides* bacteria. A Neighbour-Joining homology evolutionary tree based on the *tssH* gene confirmed the presence of T6SS in *Bacteroides* (figure 2A). Shotgun metagenomic analysis of faecal samples from the clinical GI-aGVHD and non-aGVHD groups showed higher abundance of T6SS structural genes (*tssB-tssH*) in the non-aGVHD group (figure 2B, online supplemental figure S2A, B and table S5). Higher T6SS-related gene abundance indicated better GI-aGVHD incidence and OS in allo-HSCT patients (figure 2C,D). Thus, to investigate the role of *Bacteroide* T6SS in aGVHD, a ΔT6SS mutant strain lacking *tssb*, *tssc* and *tssh* was constructed from *B. fragilis* type strain (figure 2E), following protocols in the previous study.[21] The generated mutant exhibited the same viability as WT and could express effector proteins and corresponding immune proteins but lacked the apparatus to secrete effector proteins into target cells (figure 2F, online supplemental figure S2C, D). A particular antibiotic conditioning course was applied to facilitate the clearance of residual antibiotics and indigenous microbiota,[22] resulting in solid *Bacteroide* transplantation, observed after 7 days of bacteria administration (online supplemental figure S2E). BALB/c mice were administered with *B. fragilis* WT, *B. fragilis* ΔT6SS and PBS, respectively. BALB/c recipients administrated with *B. fragilis* WT displayed higher survival rates and less pathological damage in aGVHD-targeted organs (intestine, liver and lung) compared with the control group (figure 2G–J). However, the *B. fragilis* ΔT6SS strain did not mitigate aGVHD (figure 2H). Notably, mice in the *B. fragilis* WT group displayed significantly improved intestine pathology, with intact crypt cells, reduced cystic cells and less inflammatory cell infiltration, while *B. fragilis* ΔT6SS did not have this effect (figure 2J). Mucus plays a pivotal role in the pathogenesis of aGVHD, and mucin-degrading microbes may influence disease progression by compromising the intestinal barrier and modulating immune signalling. To explore this, we performed PAS staining on intestinal tissues collected from recipient mice treated with *B. fragilis* WT, *B. fragilis* ΔT6SS or PBS control (online supplemental figure S2F). The results revealed that *B. fragilis* WT, which alleviated aGVHD in the mouse model, preserved the intestinal mucus layer more effectively than either the ΔT6SS mutant or PBS control. Taken together, these results suggest that T6SS is critical for *B. fragilis* against aGVHD in mice.

T6SS is essential for *B. fragilis* inhibition of T cell activation and protection of the intestinal barrier

Acute GVHD development involves donor T cell activation and subsequent damage to target organs. 2 weeks post-HSCT, T cell activation was assessed in aGVHD mice. The frequency of $CD4^+$ T, $CD8^+$ T cells and the $CD4^+/CD8^+$ T cell ratio in recipient spleens were comparable across the three groups (online supplemental figure S3A). Administration of *B. fragilis* WT strain significantly decreased the frequency of $CD69^+$ $CD4^+$ T cells and $CD69^+$ $CD8^+$ T cells in recipient spleens (figure 3A). Similar immune phenotypes were observed in the intestine, mainly manifested as reduced frequency of $CD69^+$ $CD4^+$ T cells in the *B. fragilis* WT group (figure 3B,C). Moreover, DC activation, indicated by $MHCII^+$ and $CD86^+$, was also significantly reduced in recipient spleens from the *B. fragilis* WT group (online supplemental figure S3B), suggesting the role of T6SS in modulation of host innate inflammation.

Intestinal tissues from recipient mice were stained with TUNEL and KI67 respectively. As a result, *B. fragilis* WT, instead of *B. fragilis* ΔT6SS, reduced cell death and improved cell proliferation in the intestine (figure 3D,E). Intestinal barrier integrity assessed using a FITC-dextran assay[23] showed that *B. fragilis* WT, but not *B. fragilis* ΔT6SS, significantly reduced serum FITC-dextran levels, indicating a more intact intestinal barrier (figure 3F). Tight junction protein expression was measured in large intestine samples 14 days post-HSCT. Both qPCR and immunofluorescence revealed higher expression of tight junction proteins in the *B. fragilis* WT group compared with control and *B. fragilis* ΔT6SS groups (figure 3G,H). Additionally, the administration of *B. fragilis* WT reduced concentrations of proinflammatory IL-1β and IL-6. The anti-inflammatory function of *B. fragilis* disappeared in the *B. fragilis* ΔT6SS mutated strain (figure 3I). Together, these results indicate that *B. fragilis* T6SS contributes to maintaining intestinal barrier maintenance by upregulating tight junction proteins and reducing intestinal inflammatory response, thereby alleviating aGVHD.

T6SS-mediated antagonism regulates BA metabolism

The structure and proportion of intestinal flora are closely associated with the occurrence and development of aGVHD. Since T6SS is known to mediate interspecies competition, 16S ASV analysis was conducted to assess how *B. fragilis* T6SS affects recipient gut microbiota in aGVHD mouse model at 14 days post-transplantation (figure 4A). Although mutation of T6SS did not affect overall microbial diversity, PCoA analysis revealed a clear separation of faecal samples collected from the WT group compared with those from the PBS group or *B. fragilis* ΔT6SS group (figure 4B). LEfSe analysis (LDA score >3.5, $p<0.05$) was used to rank the ability of different taxa to discriminate between the *B. fragilis* WT group and the *B. fragilis* ΔT6SS group (figure 4C). At phylum level, *B. fragilis* WT significantly increased the relative abundance of *Bacteroidota* and *Proteobacteria*, and *B. fragilis* ΔT6SS increased the relative abundance of *Firmicutes* and *Verrucomicrobiota* (figure 4C). At the genus level, the *B. fragilis* WT was able to increase the abundance of certain anti-inflammatory bacteria like *Lachnoclostridium Lachnospiraceae_NK4B4_group* and *Lachnospiraceae_UCG-006_group*, whereas *B. fragilis* △T6SS could not. Meanwhile, the *B. fragilis* WT was able to suppress the abundance of certain inflammatory bacteria like *Ruminococcus* (online supplemental figure S4A, B). Nevertheless, *B. fragilis* WT reduced the relative abundance of *Bacteroide* genus, but increased the total abundance of *Bacteroidota* bacteria, possibly due to T6SS's ability in intraspecies antagonism.

Microbial metabolites directly affect intestinal barrier function. Metabolomic analysis showed significant differences in the faecal metabolome between the *B. fragilis* WT and *B. fragilis* ΔT6SS groups (figure 4D). Compared with *B. fragilis* ΔT6SS, 219 metabolites differed significantly in the *B. fragilis* WT group, with 158 upregulated and 61 downregulated (figure 4E). Temporal trends in metabolite abundance were further assessed via K-means clustering, revealing three distinct profiles (figure 4F). KEGG pathway analysis highlighted that primary BA biosynthesis was the most affected pathway (figure 4F). Indeed,

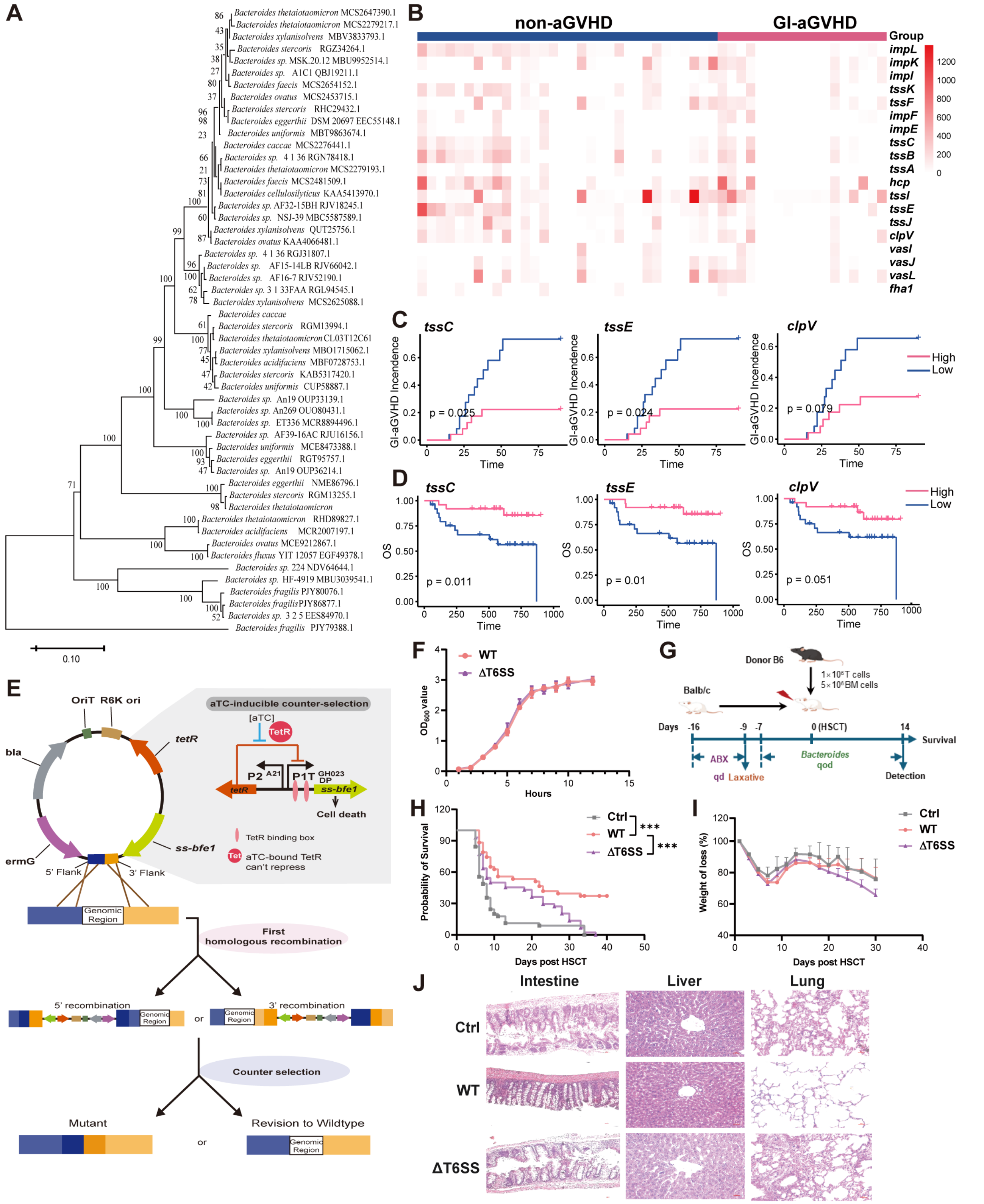

Figure 2 T6SS is required for *Bacteroides fragilis* protection on GVHD. (A) Neighbour-Joining homology evolutionary tree constructed using the *tssH* genes across *Bacteroidetes* bacteria. (B) Heatmap of T6SS-related genes in clinical samples at 4–6 weeks post-allo-HSCT. n=18 for GI-aGVHD group; n=32 for non-aGVHD group. (C) KM curves of T6SS-related gene abundance on GI-aGVHD incidences. (D) KM curves of T6SS-related gene abundance on overall survival. (E) Schematic of ΔT6SS construction using suicide vector. (F) Viability curves of *B. fragilis* strains in liquid medium. (G) Schematic illustration of mouse aGVHD model. Recipient mice (BALB/c) were gavaged with 300 µL antibiotic water for 7 days and water containing laxative for 2 days and were gavaged with the *B. fragilis* strains at a dose of 2×10^9 (CFU) for 7 days, followed by BMT. (H–I) Survival and weight loss of recipient mice transplanted as shown in (G). (J) HE stain of intestine, liver and lung in recipient mice at 14 days post-transplantation. Survival curves were analysed with the log-rank test. ***p<0.001. aGVHD, acute graft-versus-host disease; BMT, bone marrow transplantation; GI, gastrointestinal; GRFS, GVHD-free, relapse-free survival; HSCT, haematopoietic stem cell transplantation; T6SS, type VI secretion system.

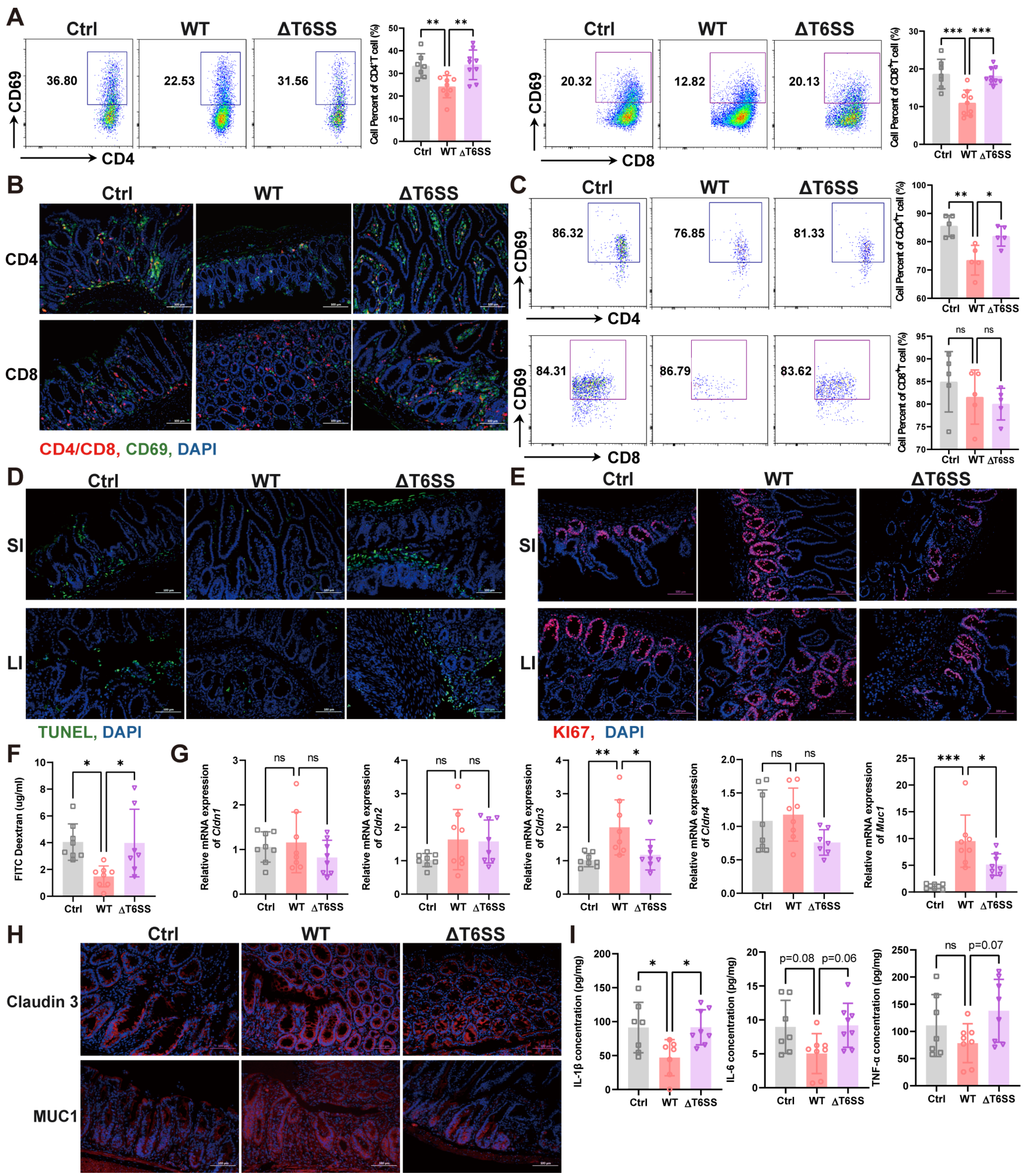

Figure 3 T6SS is essential for *Bacteroides fragilis* inhibition of T cell activation and intestinal barrier protection. (A) Representative images illustrating the FACS analysis gating strategy, and statistical chart on the spleen. n=7 for Ctrl group; n=9 for WT and T6SS group. (B) Representative microscopy images illustrating CD4^{+} and CD8^{+} T cells activation. (C) Representative images illustrating the FACS analysis gating strategy, and statistical chart on the intestine. n=5 for each group. (D, E) Representative images of TUNEL and Ki67 staining of recipient intestine. (F) Level of dextran concentration among three groups. n=8 for Ctrl and WT group; n=7 for △T6SS group. (G, H) qRT-PCR (G) and immunofluorescence (H) indicate the tight junction protein expression of recipient colonic tissue. n=8 for each group. (I) Level of inflammation cytokines in recipient colonic tissue. Tissues and organs were collected at 2 weeks post-transplantation. Data are represented as mean±SD. Each dot represents a single biological replicate. P values were analysed with the one-way ANOVA test. *p<0.05, **p<0.01, ***p<0.001. ANOVA, analysis of variance; T6SS, type VI secretion system.

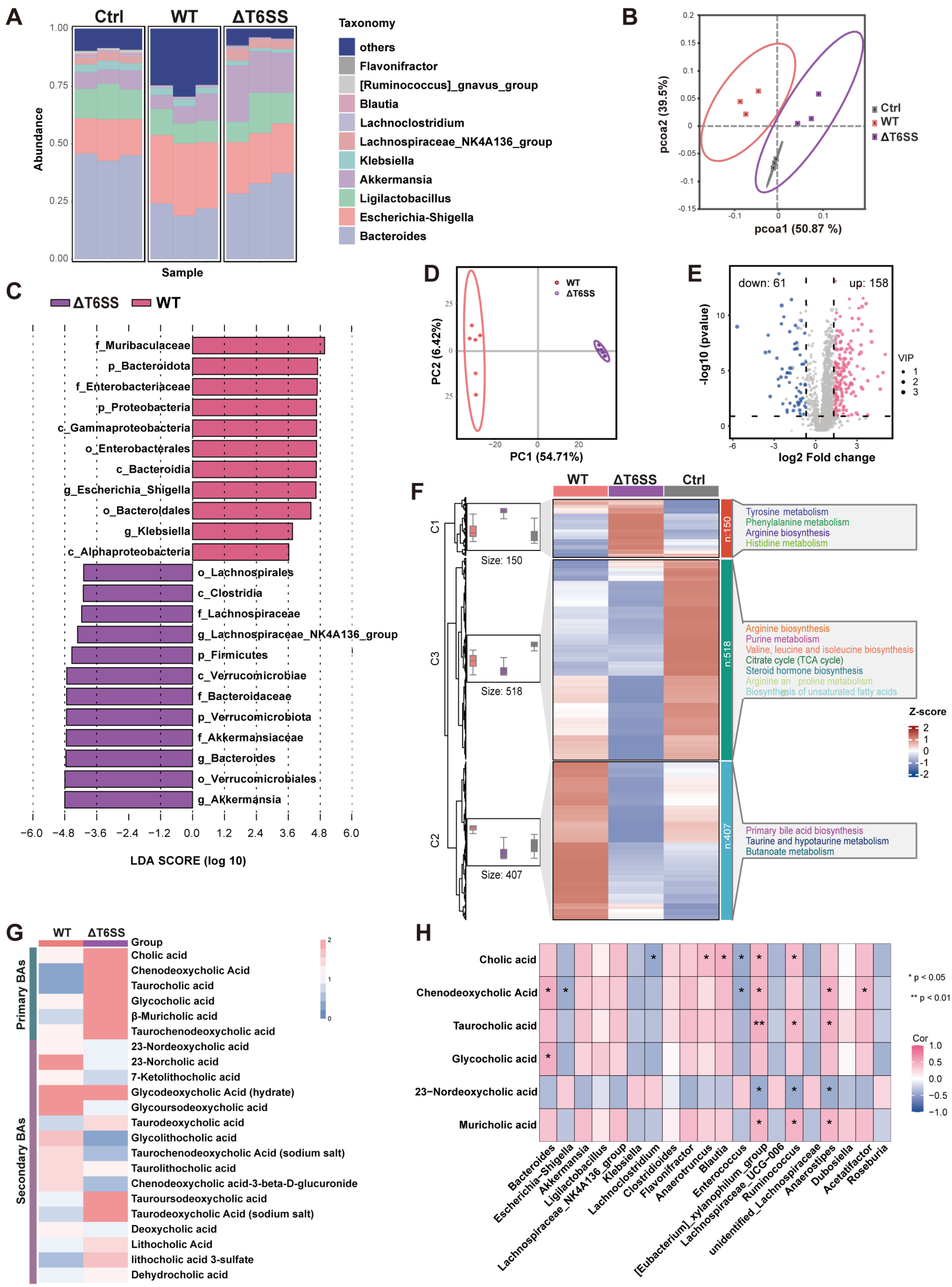

Figure 4 T6SS-mediated antagonism regulates bile acid (BA) metabolism. At 14 days post-BMT, faecal samples of recipient mice were collected for 16S rDNA sequencing and untargeted metabolism analysis. (A) Ternary plot for recipient mice's top 10 intestinal taxa. (B) PCoA analysis of intestinal microbiota structure in recipient mice. n=3 for each group. (C) LEfSe analysis on significantly different taxonomic compositions. (D) PCoA analysis of intestinal metabolome in recipient mice. n=6 for WT and ΔT6SS group. (E) Volcano plot displaying altered metabolites in recipient mice. (F) K-means clustering and KEGG enrichment analysis of different expressed metabolites. (G) Heatmap showing the relative abundance of primary BAs and secondary BAs in recipient mice. (H) Spearman analysis between BAs and altered taxa. *p<0.05, **p<0.01. BMT, bone marrow transplantation; LDA, Linear discriminant analysis; LEfSe, linear discriminant analysis effect size; PCoA, principal coordinates analysis; T6SS, type VI secretion system.

an accumulation of primary BAs was observed in the ΔT6SS group (figure 4G, online supplemental figure S4C). The alteration of BAs is positively associated with T6SS-mediated taxa change instead of the relative abundance of *B. fragilis* strains (figure 4H). These results suggest that T6SS-mediated microbiota alteration strongly affects intestinal BA metabolism.

Accumulation of primary BAs exacerbates aGVHD

BAs are synthesised in the liver and converted into secondary BAs by intestinal bacteria after being excreted into the intestine. In the mouse aGVHD model, the BA composition in *B. fragilis* recipients was distinct from that in the control and *B. fragilis* ΔT6SS groups. To assess the accumulation of primary BA on aGVHD progression, we constructed a mouse aGVHD model and administered BALB/c mice with chenodeoxycholic acid (CDCA) and methylcellulose as a control (figure 5A). As a result, the administration of CDCA significantly shortened recipients' survival time (figure 5B), due to the activation of T cells (figure 5C,D) and the loss of the intestinal barrier integrity (figure 5E–I). In addition, CDCA could promote T cell activation directly in vitro (online supplemental figure S5). These results suggest that the accumulation of primary BA exacerbates aGVHD in mouse models.

T6SS-mediated BA alteration affects GVHD outcomes

We further carried out microbial metabolome profiling in clinical faecal samples to investigate the alteration of the BAs during GI-aGVHD development. According to KEGG pathway analysis, primary BA metabolism showed the most striking change in faecal samples before and after allo-HSCT (figure 6A). Over the course of allo-HSCT, there was an increase in primary BAs in all patients, while this phenotype was more severe in patients subsequently developed GI-aGVHD (online supplemental figure S6A). Accordingly, we observed an overall reduction of bile-salt hydrolase (*bsh*) abundance in non-aGVHD patients (mean=1628 at Pre-HSCT and mean=814 at phase 3), while the reduction was more severe in patients who developed GI-aGVHD (mean=1221 at Pre-HSCT and mean=34 at phase 3) (online supplemental figure S6B). At 4–6 weeks post-allo-HSCT, the abundance of the primary BAs, but not secondary BAs, increased in GI-aGVHD patients (figure 6B). Specific primary BAs, including CDCA and tauro-β-muricholic acid (tau-β-MCA) in the GI-aGVHD group were significantly higher than in the non-aGVHD group. Consistently, the higher abundance of CDCA and tau-β-MCA was positively correlated with aGVHD risk (figure 6C,D).

In order to investigate the role of T6SS conserved genes on the occurrence of aGVHD through metabolites, the mediation package of R language was used for analysis (figure 6E). The mediating effect of T6SS-related gene families (K11889-K11919) on aGVHD risk was manifested as an indirect effect, direct effect, and total effect mediated by primary BAs (figure 6F). Specifically, gene families K11891, K11895, K11905 and K11906 were shown to reduce aGVHD risk directly. In all models, the indirect effect is significant ($p<0.05$), but its proportion in the total effect varied. The proportion of mediating effects of K11895 and K11906 was relatively high, indicating that primary BAs play an essential intermediary role in these gene families (figure 6G, online supplemental figure S7A). The abundance of K11891, K11895, K11905 and K11906 was negatively correlated with primary BAs (online supplemental figure S7B). These findings suggest that modulating primary BAs could enhance the role of T6SS gene families in reducing the risk of aGVHD.

To further assess the potential confounding impact of mucin-degrading bacteria in our mediation model (T6SS → BAs → aGVHD), we curated a panel of representative mucin-degrading taxa based on a published reference[24] and incorporated them as covariates in the mediation analysis. As shown in the online supplemental figure S7C, the core mediation effects involving T6SS-related genes, BAs and aGVHD remained robust, with only minor deviations observed after adjusting for these mucin-degrading microbes. These findings suggest that although mucus-associated bacteria may have a modest modulatory effect on the T6SS–BA–aGVHD pathway, their influence is overshadowed by the dominant regulatory role of T6SS-related mechanisms. Additional mediation analysis showed that both microbiota diversity and *bsh* abundance play a significant mediating role in T6SS-regulated primary BA levels (online supplemental figure S7D). This highlights the intricate interplay between mucin-degrading microbiota and host-microbe interactions in GVHD pathophysiology, while reinforcing the centrality of the T6SS-BA axis identified in our study.

DISCUSSION

Patients with haematological malignancies and other oncological conditions are particularly vulnerable to increased intestinal permeability due to the frequent use of broad-spectrum antibiotics and chemoradiotherapy.[25 26] FMT, or targeted probiotic supplementation, has been proposed as a clinical strategy to restore intestinal homeostasis.[27] However, clinical application remains challenging, especially in immunocompromised patients such as severely ill children and premature infants, where the risk of bacterial translocation or antibiotic-resistant bacteria poses significant concerns.[28–30] Thus, a deeper understanding of the interaction between gut micro-organisms and the intestinal barrier is crucial to identify novel bacterial products that can be harnessed for therapeutic use. In this context, our study highlights the role of *B. fragilis*, a key intestinal commensal, in regulating gut microbiota composition through its T6SS. We demonstrate that the T6SS of *B. fragilis* not only modulates microbial community dynamics but also significantly impacts the metabolome, particularly BA metabolism, thereby improving intestinal barrier integrity and mitigating the progression of aGVHD. The T6SS-BA axis merged as a crucial internal mechanism for maintaining long-term intestinal stability, positioning it as a potential target for therapeutic intervention.

We compared the gut microbiota of GI-aGVHD patients to those without aGVHD after allo-HSCT, confirming previous findings on the protective role of *Bacteroides* spp in aGVHD. Our analysis further identified T6SS as a critical target for microbial intervention. While initially perceived as a bacterial antagonism mechanism in gram-negative bacteria, recent research has expanded the understanding of T6SS.[14 17] In enteric pathogens, T6SS contributes to pathogenesis by promoting host-pathogen interactions, whereas in commensals, T6SS helps secure ecological niches by outcompeting other bacteria. In the mouse aGVHD model, administration of *B. fragilis* reduced donor T-cell activation, improved intestinal barrier function by tight junction protein expression, and decreased intestinal inflammation, leading to an improved survival rate in aGVHD models. In contrast, a T6SS-paralysed mutant lost these protective effects, hinting at the pivotal role of T6SS in maintaining intestinal homeostasis during aGVHD.

The GI tract is a primary target for alloreactive T cells following allogeneic BMT.[31 32] Intestinal damage promotes the release of proinflammatory cytokines and activates host

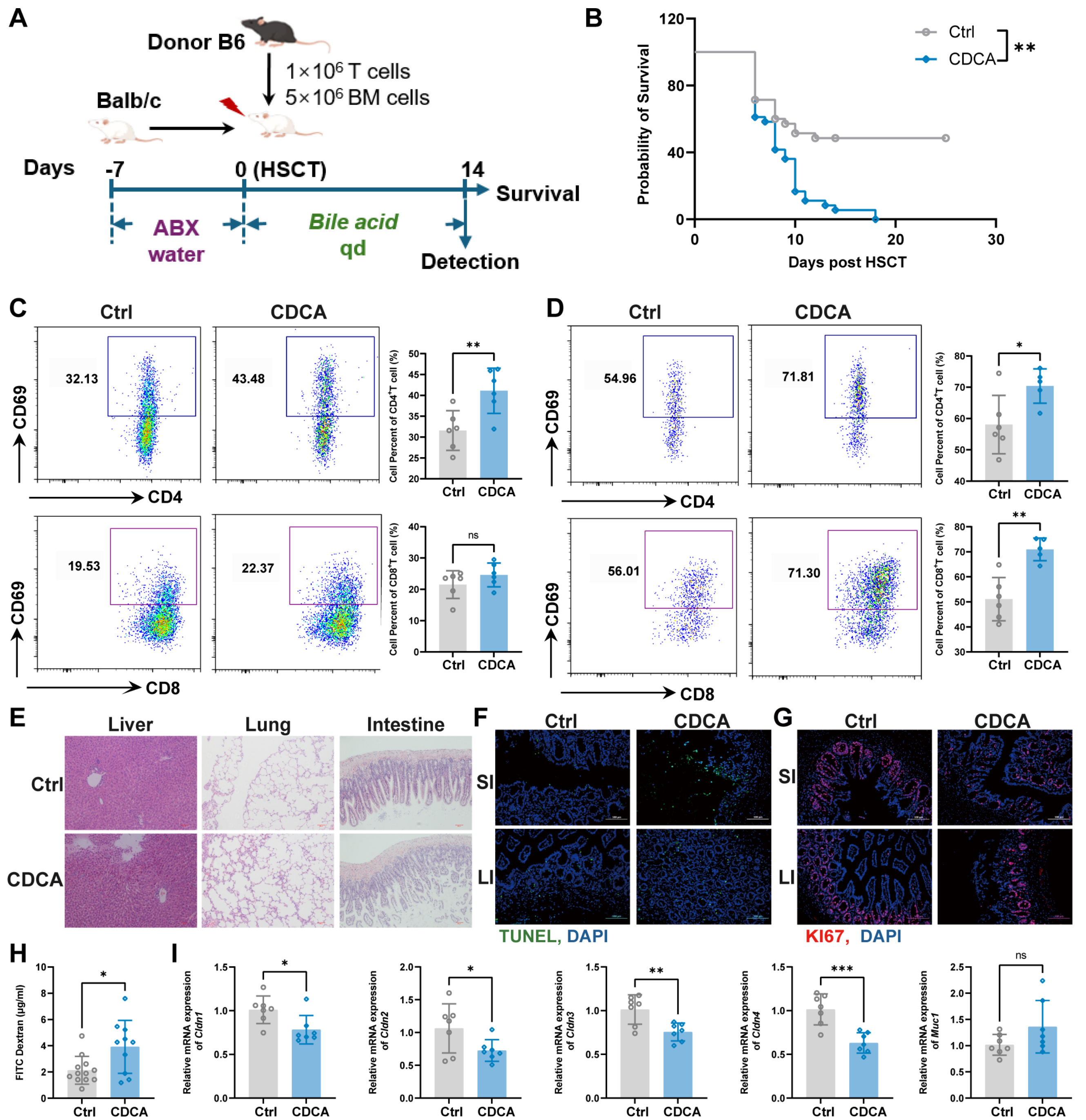

Figure 5 Accumulation of primary bile acids exacerbates aGVHD. (A) Schematic illustration of mouse transplantation model. Recipient mice were gavaged with 300 µL antibiotic water for 7 days, followed by BMT. Recipient mice were gavaged with CDCA (100 mg/kg) for 14 days. (B) Survival of recipient mice transplanted as shown in (A). (C, D) Percentage of activated T cells in spleen (C) and intestine (D). n=5–6 for each group. (E) HE stain of liver, lung and colonic tissues in recipient mice at 2 weeks post-transplantation. (F, G) Representative images of TUNEL and Ki67 staining of recipient intestine. (H) Level of dextran concentration among three groups. n=12 for Ctrl group, n=10 for CDCA group. (I) qRT-PCR indicates the tight junction protein expression of recipient colonic tissue. n=7 for each group. Data are represented as mean±SD. Each dot represents a single biological replicate. P values were analysed with the Student's t-test. *p<0.05, **p<0.01, ***p<0.001. aGVHD, acute graft-versus-host disease; BMT, bone marrow transplantation; HSCT, haematopoietic stem cell transplantation.

antigen-presenting cells, which in turn amplify the presentation of host antigens to donor T cells, exacerbating aGVHD. Increased intestinal permeability allows bacteria and toxins to break the mucosal barrier, compromising intestinal integrity. Our findings indicated that *B. fragilis* applied T6SS for interspecific competition and reduced the abundance of bacteria of the same genus, but the final result of competition was an increase in the overall abundance of *Bacteroidota*. This further proved that the intestine is a complex antagonistic system, as the vacancy of certain species will cause adjacent species to quickly occupy the ecological niche and reproduce. T6SS-regulated species at the phylum level further affect the overall metabolome, together contributing to intestinal homeostasis.

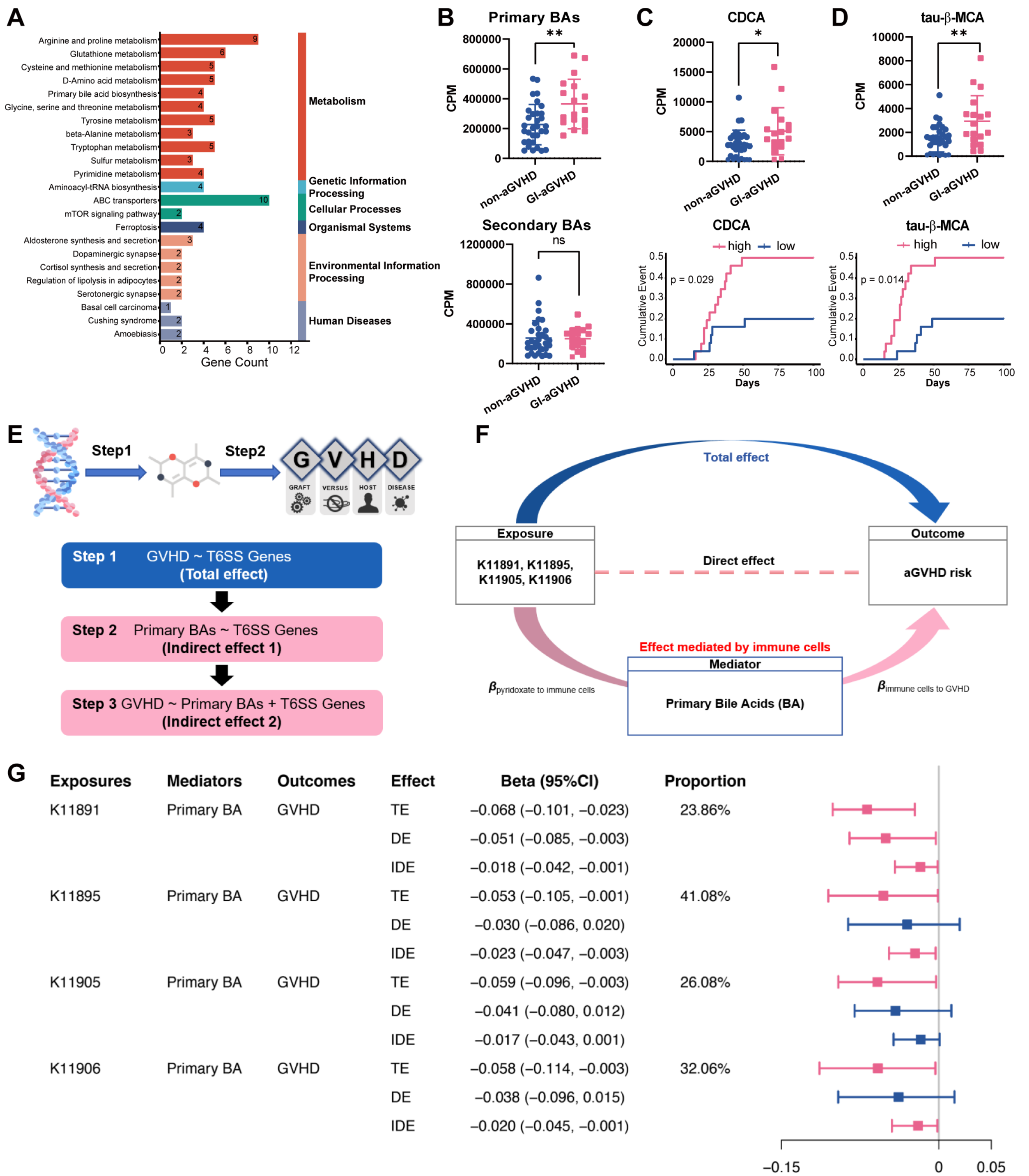

Figure 6 T6SS-mediated BA alteration affects GVHD outcomes. (A) Differentially expressed metabolites in clinical faecal samples at 4–6 weeks post-HSCT were selected and enriched in KEGG pathway. (B) Relative abundance of primary BAs and secondary BAs in faecal samples at 4–6 weeks post-HSCT. (C, D) Relative abundance of CDCA and tau-β-MCA and corresponding KM curves with aGVHD risk. (E) Schematic illustration of mediation analyses. (F, G) The proportion of mediating effects of T6SS-related genes on aGVHD risk. *p<0.05, **p<0.01. aGVHD, acute graft-versus-host disease; GI, gastrointestinal; HSCT, haematopoietic stem cell transplantation; tau-β-MCA, tauro-β-muricholic acid; T6SS, type VI secretion system.

Independent studies have demonstrated that BA metabolism is significantly altered in patients with aGVHD. BAs exert regulatory effects on intestinal homeostasis through BA receptor signalling.[33–35] Consistent with previous studies, this study provided supporting evidence that BA metabolism, especially the accumulation of primary BAs, plays an important regulatory role in the development of aGVHD.[33 34] BA metabolism initiates in the liver (via CYP7A1 and CYP27A1) and produces primary BAs, which are further processed by gut microbes into secondary BAs. Secondary BAs, such as lithocholic acid and ursodeoxycholic acid, promote epithelial regeneration, regulate mucus layer formation and prevent intestinal inflammation.[36–38] BAs can alter the diversity of gut microbiota composition and provide sufficient energy to support the diversity of many micro-organisms.[39 40] In turn, the intestinal micro-organisms can regulate BA synthesis through the FXR-FGF15/19 feedback mechanism, shaping intestinal stability.

The BA pool is a potential candidate for identifying bacterial products that can regulate intestinal homeostasis. Recent research has expanded the functions of secondary BAs to much broader areas, including affecting Treg/TH17 cell differentiation, macrophage polarisation, and inhibition of NLRP3 inflammation.[41–43] Altered BA metabolism was confirmed to exacerbate T-cell-driven inflammation during GVHD.[33] However, the underlying mechanisms of BA imbalance in aGVHD are still in the initial stages of exploration. In this study, we preliminarily identified correlations between specific bacterial species and BA alterations, suggesting that T6SS-mediated interactions with BA metabolism could be an essential factor in this imbalance. Whether the accumulation of primary BA is due to the paralysed metabolism capacity of intestinal bacteria or the feedback by interacting with intestinal BA receptors requires in-depth mechanism research and functional verification. Further research is required to elucidate the precise metabolic pathway governing secondary BA synthesis and to explore their potential as therapeutic targets for aGVHD management.

Gut microbiota remains stable throughout a healthy person's lifetime but dramatically changes with factors such as diet, medication and various diseases. In a healthy microbiome, strains with functional redundancy coexist, ensuring resilience against external perturbations. However, when disruptions occur, they can lead to unpredictable community-level changes, affecting overall health.[44] Understanding the mechanisms that maintain microbial stability is crucial for developing effective therapeutic interventions. T6SS, which can shape the gut microbiota composition, offers a promising strategy for restoring and maintaining microbial homeostasis. Advances in biotechnology have enabled the construction of engineered suppressor cells that express nanobodies targeting specific bacterial strains, enhancing the killing capacity and spectrum of T6SS.[45] These developments hold significant potential for precision medicine, allowing for the targeted modulation of gut bacteria to treat various diseases, including aGVHD. Further exploration of T6SS in commensal bacteria will deepen our understanding of the molecular interactions between gut micro-organisms and human cells, paving the way for safer and more effective microbial-based therapies in clinical practice.

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